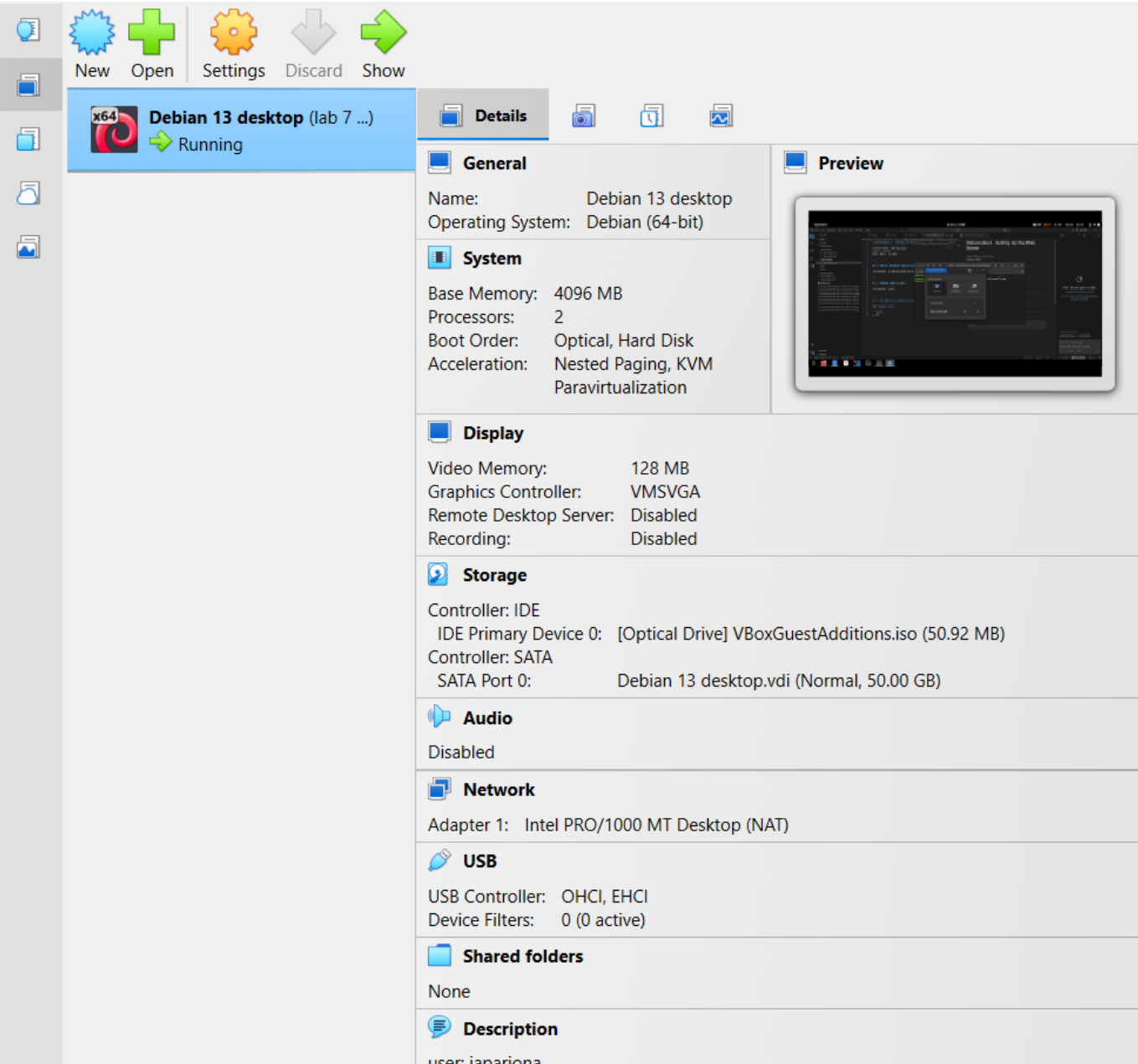


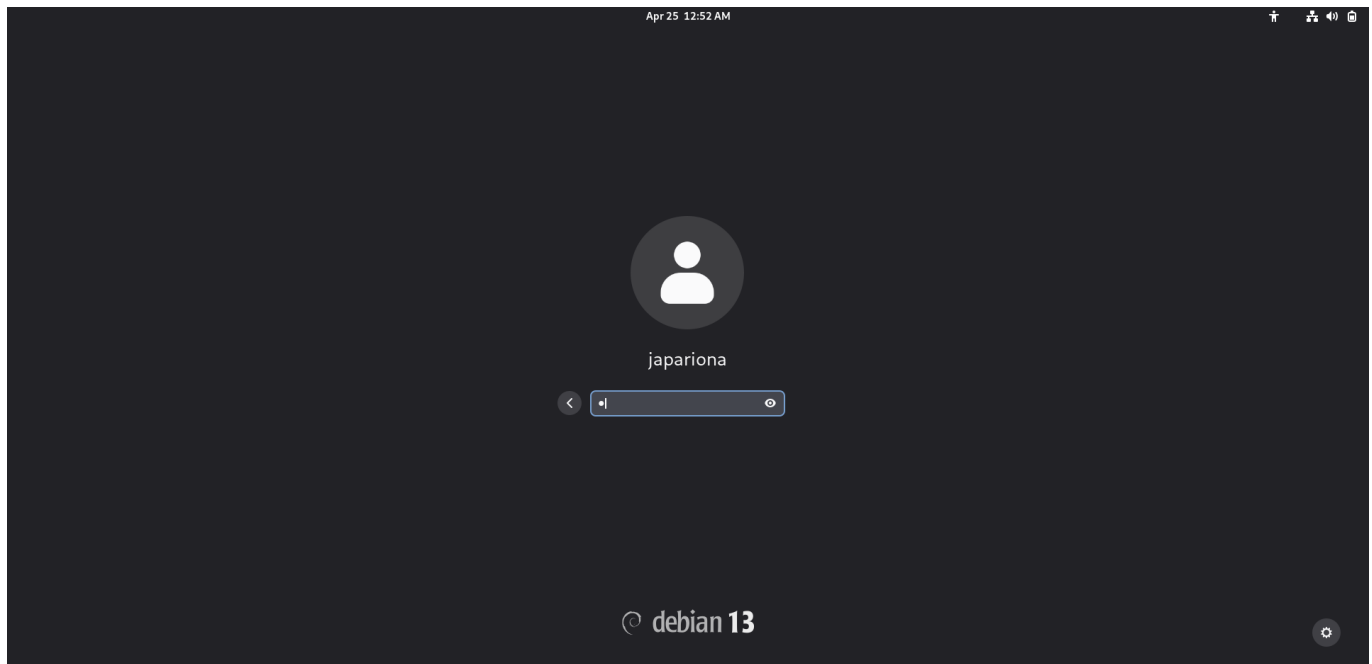
# Deliverable 2 - Setting Up The Web Server

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Course: CIS106  
Date: April 25 2026

## 1. Server Hardware Specifications



## 2. Debian Login Screen



### 3. IP Address of Debian Server

```
Tilix: japariona@cis106: ~  
1: japariona@cis106: ~  
japariona@cis106:~$ hostname -I  
10.0.2.15 fd17:625c:f037:2:5af0:ab48:e145:17e4 fd17:625c:f037:2:a00:27ff:fe8d:cda8  
japariona@cis106:~$
```

#### Command Used

```
hostname -I
```

### 4. How do you work with the Firewall in Debian? (Type and explain what each command does)

#### a. How do you check if the Firewall is running?

This command checks the current status of UFW (Uncomplicated Firewall). It lets me know if the firewall is active, inactive, and can also show the rules that are currently applied to the system. This is useful to verify if my server protection is on.

```
sudo ufw status
```

## b. How do you disable the Firewall?

This command disables the firewall and stops it from filtering incoming or outgoing traffic. I would normally use this for testing, troubleshooting connection problems, or if I need to temporarily turn it off.

```
sudo ufw disable
```

## c. How do you add Apache to the Firewall?

This command creates a rule that allows Apache web server traffic through the firewall. It opens the needed web ports so users can access the hosted website from a browser.

```
sudo ufw allow 'Apache'
```

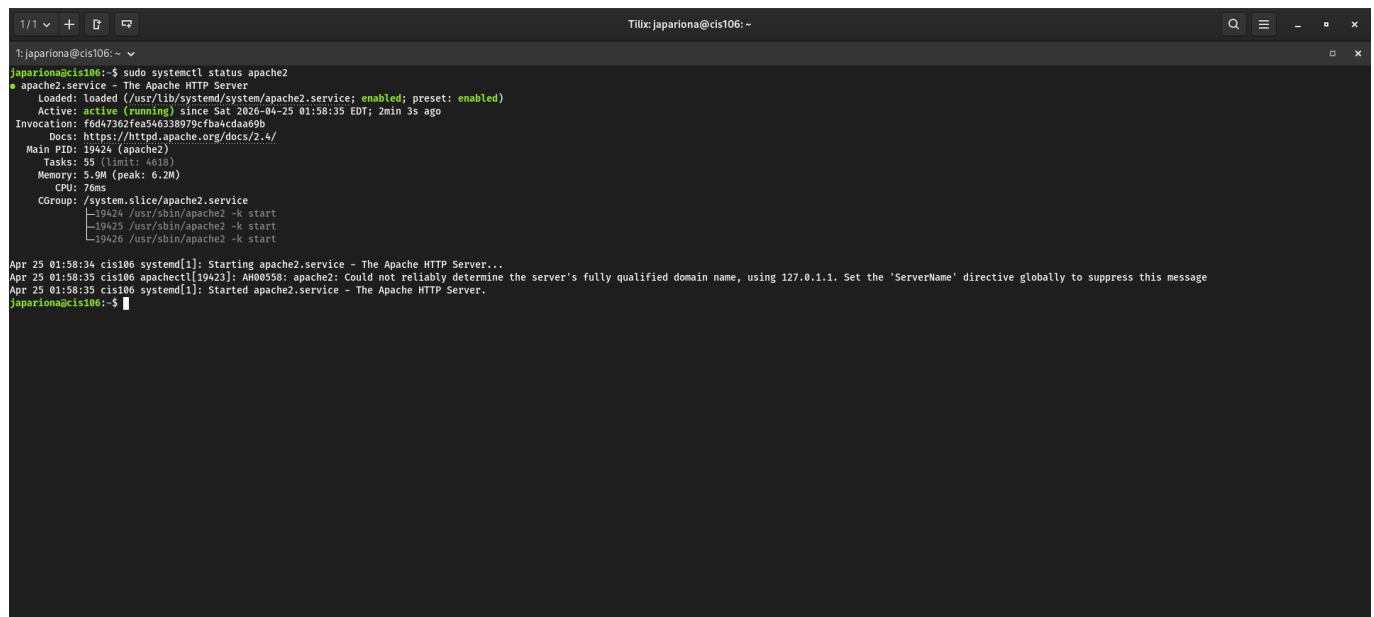
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# 5. What different commands do we use to work with Apache?

## 1. What is the command you use to check if Apache is running?

This command checks the current status of the Apache service. It shows if Apache is active, running, stopped, or if there are any recent service messages. This is one of the main commands I would use to make sure the web server is working properly.

- The command is: `sudo systemctl status apache2`

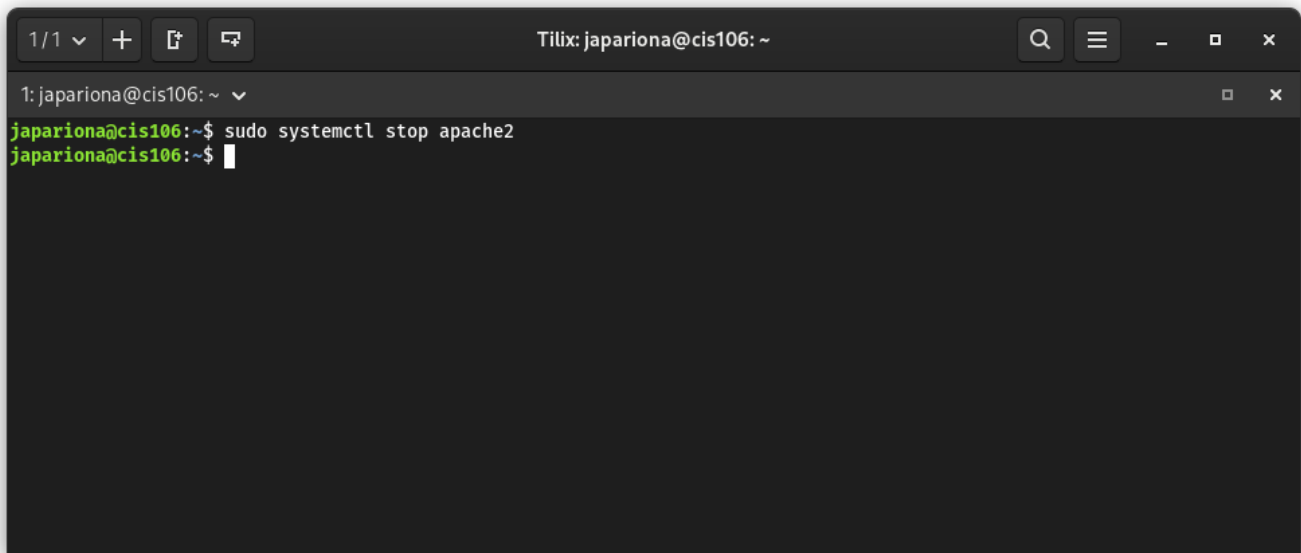


```
1:japariona@cis106:~  
japariona@cis106:~$ sudo systemctl status apache2  
● apache2.service - The Apache HTTP Server  
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)  
   Active: active (running) since Sat 2026-04-25 01:58:35 EDT; 2min 3s ago  
     Invocation: fed47362fea546338979cfba4cdaa69b  
       Docs: https://httpd.apache.org/docs/2.4/  
    Main PID: 19424 (apache2)  
      Tasks: 35 (limit: 4618)  
     Memory: 5.9M (peak: 6.2M)  
        CPU: 76ms  
    CGroup: /system.slice/apache2.service  
            └─19424 /usr/sbin/apache2 -k start  
              └─19425 /usr/sbin/apache2 -k start  
                └─19426 /usr/sbin/apache2 -k start  
  
Apr 25 01:58:34 cis106 systemd[1]: Starting apache2.service - The Apache HTTP Server...  
Apr 25 01:58:35 cis106 apache2ctl[19423]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress this message  
Apr 25 01:58:35 cis106 systemd[1]: Started apache2.service - The Apache HTTP Server.  
japariona@cis106:~$
```

## 2. What is the command you use to stop Apache?

This command stops the Apache web server service completely. I would use this if I need to restart it later, make changes safely, or shut the web server down for maintenance.

- The command is: `sudo systemctl stop apache2`

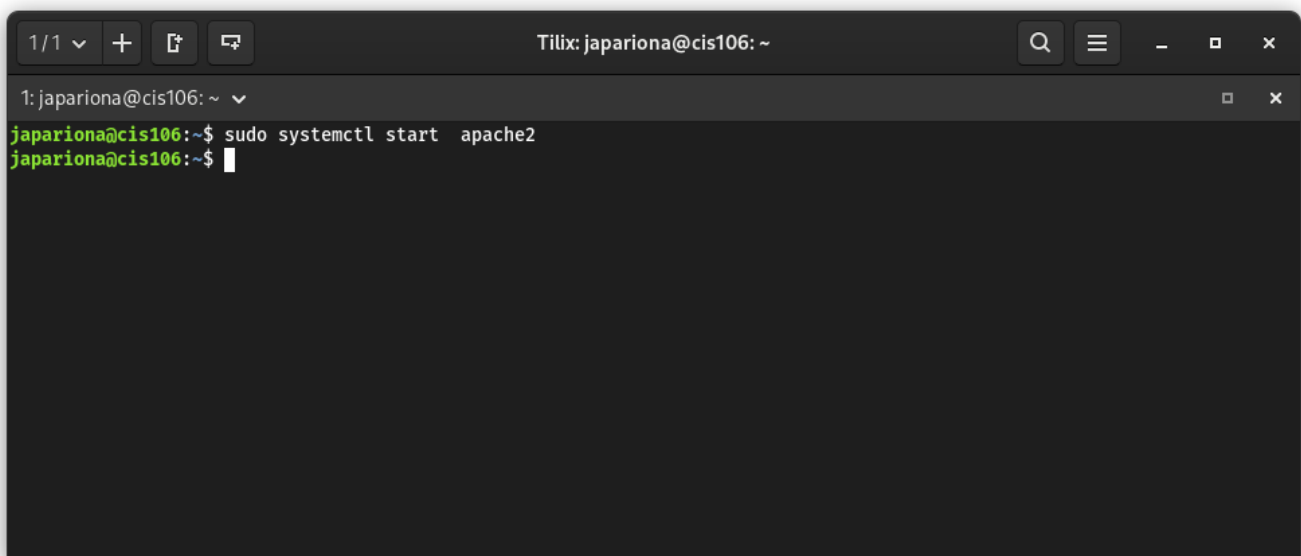
A terminal window titled 'Tilix: japariona@cis106: ~' showing a command prompt. The user enters 'sudo systemctl stop apache2' and the prompt returns. The terminal has a dark background with green and yellow text for the prompt and command respectively.

```
1: japariona@cis106: ~  
japariona@cis106:~$ sudo systemctl stop apache2  
japariona@cis106:~$
```

### 3. What is the command you use to start Apache?

This command starts the Apache web server service. If Apache was stopped or the server was rebooted, this command brings the service back online so the website can be accessed again.

- The command is: `sudo systemctl start apache2`

A terminal window titled 'Tilix: japariona@cis106: ~' showing a command prompt. The user enters 'sudo systemctl start apache2' and the prompt returns. The terminal has a dark background with green and yellow text for the prompt and command respectively.

```
1: japariona@cis106: ~  
japariona@cis106:~$ sudo systemctl start apache2  
japariona@cis106:~$
```

### 4. What is the command used to test Apache configuration?

This command checks Apache configuration files for syntax errors or mistakes before restarting the service. It helps prevent bad configuration changes from breaking the web server.

- The command is: `sudo apache2ctl configtest`

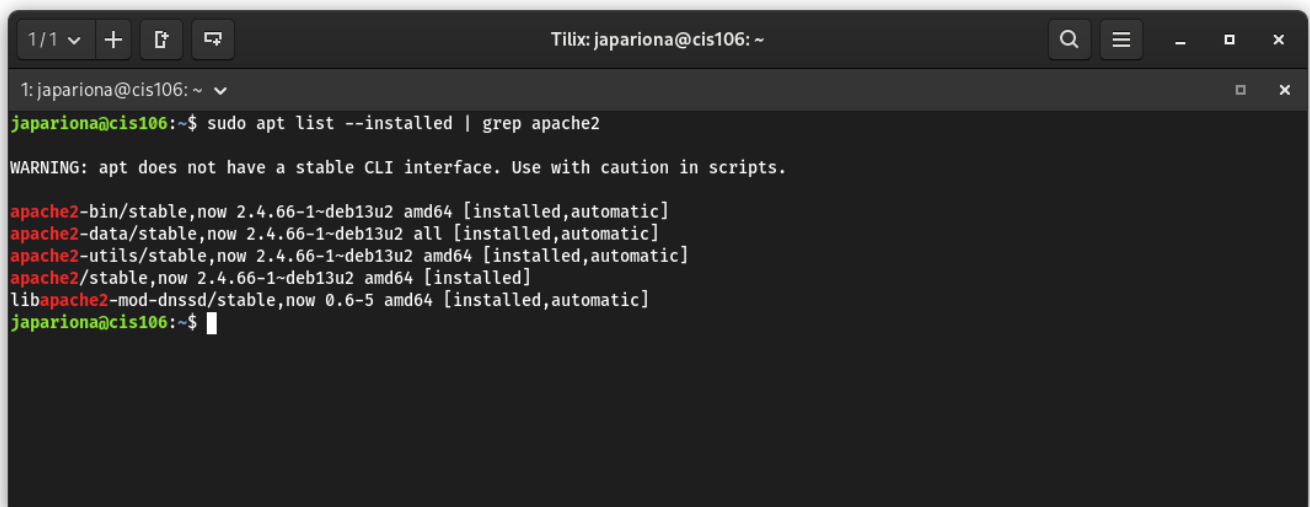


```
1/1 + [icons] Tilix: japariona@cis106: ~
1: japariona@cis106: ~
japariona@cis106:~$ sudo apache2ctl configtest
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress this message
Syntax OK
japariona@cis106:~$
```

## 5. What is the command used to check the installed version of Apache?

This command searches installed packages related to Apache and shows the installed version number. It is useful when confirming what version is currently running on the system.

- The command is: `sudo apt list --installed | grep apache2`



```
1/1 + [icons] Tilix: japariona@cis106: ~
1: japariona@cis106: ~
japariona@cis106:~$ sudo apt list --installed | grep apache2
WARNING: apt does not have a stable CLI interface. Use with caution in scripts.

apache2-bin/stable,now 2.4.66-1~deb13u2 amd64 [installed,automatic]
apache2-data/stable,now 2.4.66-1~deb13u2 all [installed,automatic]
apache2-utils/stable,now 2.4.66-1~deb13u2 amd64 [installed,automatic]
apache2/stable,now 2.4.66-1~deb13u2 amd64 [installed]
libapache2-mod-dnssd/stable,now 0.6-5 amd64 [installed,automatic]
japariona@cis106:~$
```

## 6. What are some common configuration files for Apache?

This command lists the Apache configuration directory. Inside this folder are common files such as the main config file, virtual host files, ports configuration, and enabled modules.

- The command is: `ls /etc/apache2/`

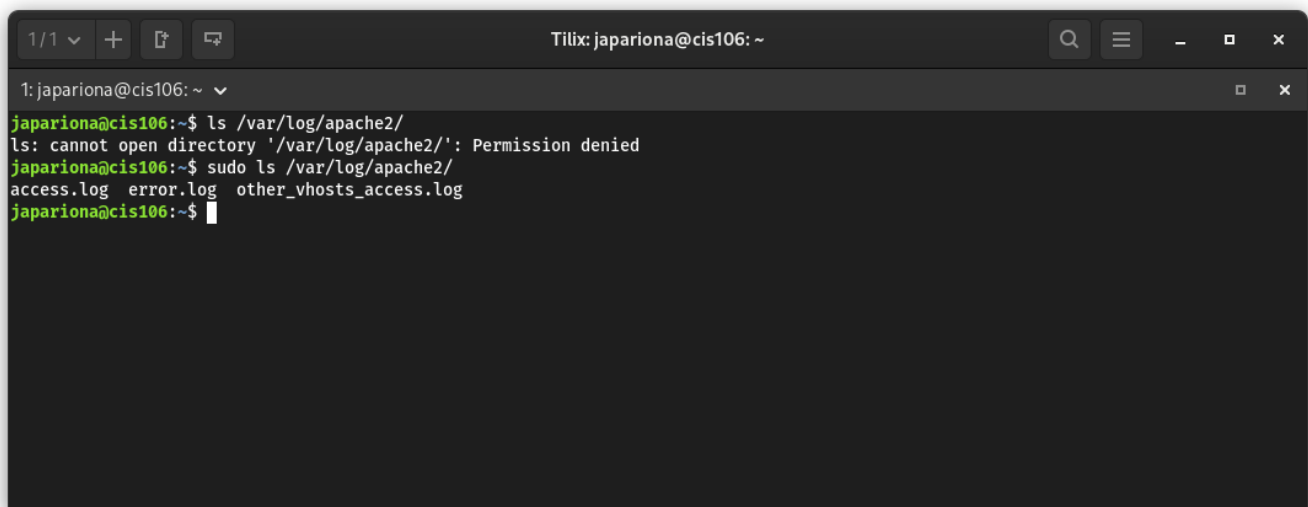


```
1/1 + [ ] [ ] Tilix: japariona@cis106: ~
1: japariona@cis106: ~
japariona@cis106:~$ ls /etc/apache2/
apache2.conf  conf-enabled  magic          mods-enabled  sites-available
conf-available  envvars      mods-available  ports.conf    sites-enabled
japariona@cis106:~$
```

## 7. Where does Apache store logs?

This command opens the Apache log directory. This is where Apache stores important files like access logs and error logs, which help track visitors and troubleshoot problems.

- The command is: `ls /var/log/apache2/`

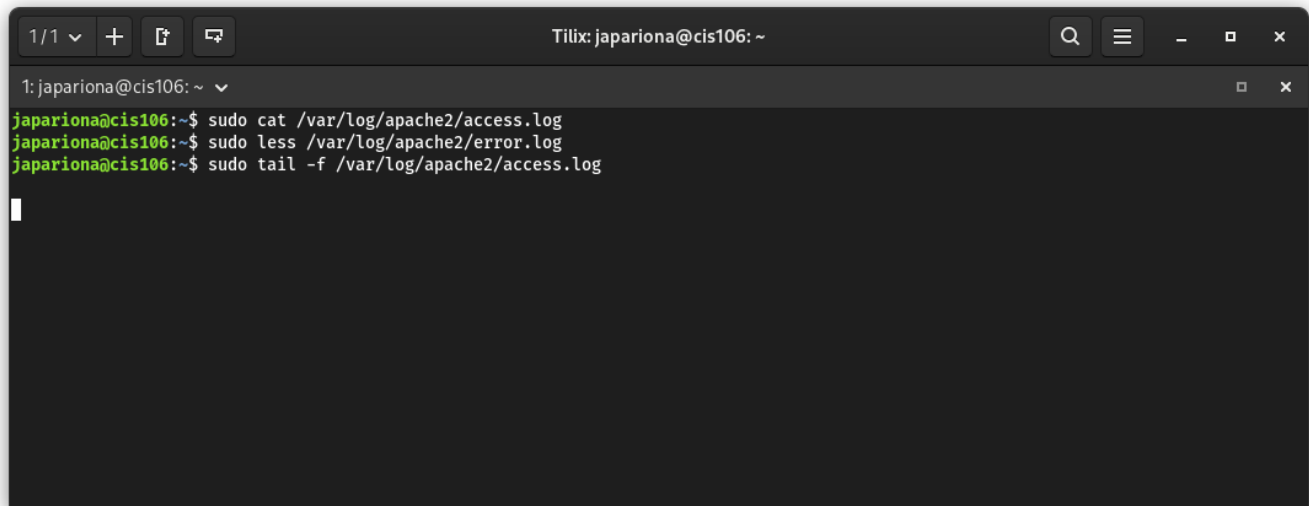


```
1/1 + [ ] [ ] Tilix: japariona@cis106: ~
1: japariona@cis106: ~
japariona@cis106:~$ ls /var/log/apache2/
ls: cannot open directory '/var/log/apache2/': Permission denied
japariona@cis106:~$ sudo ls /var/log/apache2/
access.log  error.log  other_vhosts_access.log
japariona@cis106:~$
```

## 8. What are some basic commands we can use to review logs?

These commands are used to read and monitor Apache logs. They help me check website activity, view errors, and watch new log entries in real time while testing.

- The command is: `cat /var/log/apache2/access.log`
- The command is: `less /var/log/apache2/error.log`
- The command is: `tail -f /var/log/apache2/access.log`

A terminal window titled 'Tilix: japariona@cis106: ~' with standard window controls. The terminal shows three commands entered by the user 'japariona@cis106':

```
japariona@cis106:~$ sudo cat /var/log/apache2/access.log
japariona@cis106:~$ sudo less /var/log/apache2/error.log
japariona@cis106:~$ sudo tail -f /var/log/apache2/access.log
```

The terminal is currently empty, waiting for the output of the last command.

```
1/1 v + [ ] [ ]
Tilix: japariona@cis106: ~
1: japariona@cis106: ~ v
japariona@cis106:~$ sudo cat /var/log/apache2/access.log
japariona@cis106:~$ sudo less /var/log/apache2/error.log
japariona@cis106:~$ sudo tail -f /var/log/apache2/access.log
```